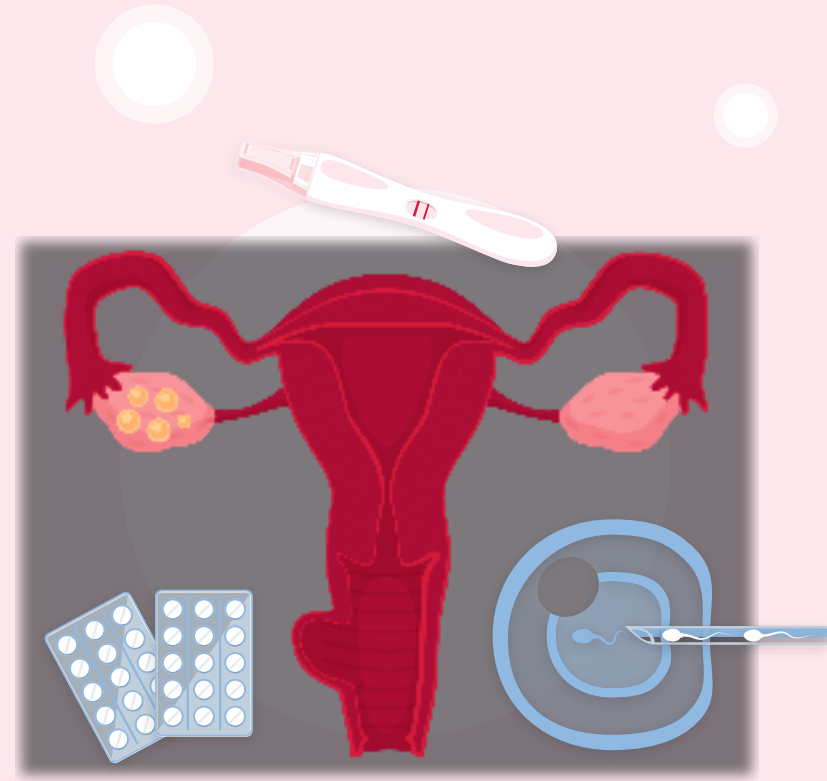


Recurrent Pregnancy loss

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OUTLINE

- ❑ **What is miscarriage & how it is different from recurrent miscarriage**
- ❑ **Categories of recurrent miscarriage**
- ❑ **Etiology of recurrent miscarriage**
- ❑ **How can evaluate and manage each etiology**





Definition of Miscarriage & RPL

Miscarriage: spontaneous loss of pregnancy before fetal viability (≤ 24 weeks)

Recurrent Pregnancy Loss (RPL) / Habitual abortion:

Defined as three or more consecutive miscarriages before 24 weeks.



Epidemiology

Incidence

- Recurrent miscarriage affects ~1% of women.
- If RM is defined as two losses, incidence doubles (~1.9%).
- up to 50% remain unexplained
 - o emotionally challenging for patients, families, and physicians



Key Determinants

- ✓ Maternal age is the strongest factor — incidence rises with age.
- ✓ Number of prior miscarriages affects future live birth chance.

New Graded Care Approach:

- ☐ After one miscarriage: give information & support.
- ☐ After two: offer initial investigations & reassurance scans.
- ☐ After three: full evidence-based evaluation per guideline.

Categories of RPL



Primary RPL

pregnancy loss in
women who have never
had a live birth

Secondary RPL

pregnancy loss in women
with at least one previous
live birth

Etiology

01

Genetic Factors

02

Thrombophilia

03

Anatomical factors

04

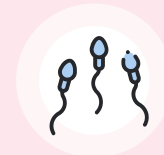
Endocrine factor

05

Immune Factors

06

Idiopathic Factors



Clinical Evaluation

General Rule

Although each cause has its own tests and management, a thorough history and physical examination are the foundation of evaluation.



History

Ask about

- Number & timing of miscarriages
- Previous pregnancies, complications
- Family history, medical conditions, lifestyle



Physical Examination

- General exam - BMI, thyroid, signs of systemic disease
- Pelvic exam - uterine size, tenderness, masses
- Speculum - cervix, infection, discharge

Genetic factor

Fetal Chromosomal Anomalies

- Most common cause of recurrent miscarriage.
- Types (in descending frequency)
 - o Trisomy
 - o Polyploidy
 - o Monosomy
 - o Others

Parental Chromosomal Rearrangements

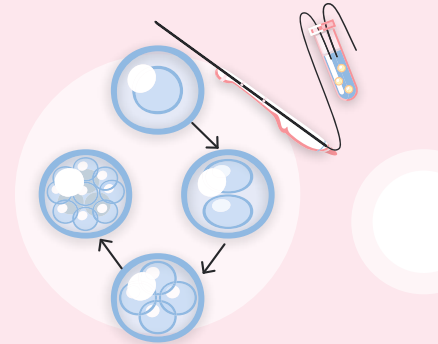
- Incidence of parental chromosomal rearrangements is significantly associated with recurrent miscarriage.
- Risk of miscarriage depends on the type of rearrangement:
 - o Reciprocal translocations
 - o Robertsonian translocations
 - o Inversions
 - o Other chromosomal anomalies

Investigation

- Alongside routine investigations, cytogenetic analysis of pregnancy tissue provides a diagnosis in >90% of couples.
- Parental peripheral blood karyotyping is recommended:
 - o When pregnancy tissue testing is unsuccessful.
 - o When no tissue is available for testing.

Management of Genetic Factors

- VF/ICSI + PGT-SR (Preimplantation Genetic Testing for Structural Rearrangements)
 - o Screens embryos for chromosomal abnormalities
 - o Transfers only healthy embryos → ↓ miscarriage risk
- Gamete Donation
 - o Option if severe parental chromosomal abnormality



Thrombophilia

Acquired — APS

- Diagnosis

Requires at least one clinical criterion and one laboratory criterion, confirmed on two occasions at least 12 weeks apart.

Clinical criteria:

1. Vascular thrombosis
2. Pregnancy morbidity includes:
 - ≥ 3 miscarriages < 10 weeks
 - ≥ 1 unexplained normal fetal loss > 10 weeks
 - ≥ 1 preterm birth < 34 weeks due to eclampsia, severe pre-eclampsia or placental insufficiency, or

Laboratory criteria:

- Lupus anticoagulant positivity
- Anticardiolipin antibody (aCL) of IgG or IgM isotype at moderate/high titer
- Anti-B2 glycoprotein I antibody of IgG or IgM isotype.

Inherited Thrombophilia

Factor V Leiden, Protein C/S deficiency, Antithrombin deficiency,
Prothrombin mutation
Cause systemic thrombosis

Management

- APS: Aspirin 75 mg + LMWH/UFH from positive test → 34 week
- Inherited thrombophilia: Routine anticoagulation not recommended

Anatomical Factors

Congenital Uterine Anomalies

- Septate uterus is the most common cause related to Recurrent miscarriage

Acquired Uterine Factors

- Myomas: Submucosal, Intramural/Subserosal
- Endometrial polyps
- Intrauterine adhesions (Asherman's syndrome)
- Adenomyosis

Investigation

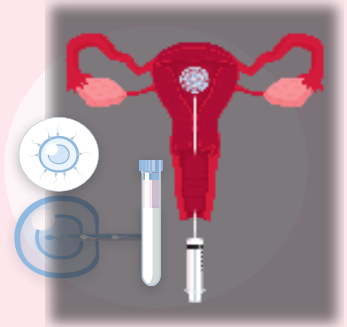
- ⁿ 3D Ultrasound
- Saline infusion sonography
- Hysterosalpingo Saline-infusion sonography (HSG) & 2D Ultrasound

Management

- ^t Uterine septum resection: Insufficient evidence to prevent recurrent miscarriage
- Open uterine surgery: Not recommended — risk of infertility and uterine scar rupture

Cervical Integrity

- Cervical insufficiency is a major cause of 2nd-trimester miscarriage, along with infection and congenital uterine anomalies
- True incidence unknown; diagnosis is primarily clinical
- Typical history: painless cervical dilation, often with intact membranes until fetal expulsion
- **Diagnosis by** Transvaginal ultrasound
- **Management by** Prophylactic vaginal progesterone and cervical cerclage



Endocrine factor

Common endocrine causes include:

- Diabetes mellitus
- Thyroid disorders
- Polycystic ovary syndrome (PCOS)
- Prolactin imbalance
- Luteal phase defect



Investigation

n

- Thyroid function tests and thyroid peroxidase (TPO) antibodies should be offered to all women with recurrent miscarriage.
 - Other endocrine assessments (e.g. for diabetes or hyperprolactinaemia) only if clinically indicated.

Treatment of Endocrine Factors

- **Thyroid Disorders:**

Correct hypo- or hyperthyroidism before conception.

- **Diabetes Mellitus:**

Maintain optimal glycemic control pre- and post-conception.

- **Polycystic Ovary Syndrome (PCOS):**

Lifestyle modification:

BMI 19–25 kg/m²

Stop smoking

Limit alcohol

Caffeine < 200 mg/day

- **Progesterone Supplementation:**

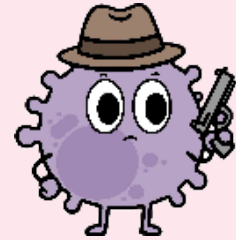
For women with recurrent miscarriage and early pregnancy bleeding.

Micronised progesterone 400 mg twice daily until 16 weeks.

Immune factors

Proposed mechanisms:

- Abnormal HLA compatibility
- Cytokine polymorphisms
- Altered natural killer (NK) cell activity
- Evidence remains inconsistent and insufficient for clinical application.
- No routine immune screening or empirical immunotherapy recommended.



NK cells

Idiopathic recurrent miscarriage

- Diagnosed when genetic, anatomical, endocrine, thrombophilic, and immune causes are excluded.
- Approximately 50% of cases remain unexplained after standard investigations.
- According to Green-top Guideline No. 17: No further investigations are required once standard evaluations are normal.



Treatment of Idiopathic Cases

- No proven pharmacological therapy.
- Aspirin, heparin, PGT-A, or hormones not recommended unless indicated.
- Supportive care in dedicated miscarriage clinic.
- Prognosis: ~75% achieve live birth in future pregnancies.
- Provide reassurance and continuous support.



CONCLUSION

- Recurrent miscarriage causes significant emotional distress.
- Counselling, reassurance, and support are essential.
- Prognosis is positive: 60–70% achieve successful pregnancy.
- Best outcomes:
 - Identify and treat causes
 - Maintain healthy lifestyle and early care
 - Provide emotional and medical support
- **Take-home message:** Empathetic care leads to successful outcomes and emotional wellbeing.

References:

1. Obstetrics – 10 Teachers' Textbook of Obstetrics, 20th Edition. Elsevier.
2. Royal College of Obstetricians and Gynaecologists (RCOG). The Investigation and Treatment of Couples with Recurrent First Trimester and Second Trimester Miscarriage. Green-top Guideline No. 17. London: RCOG; 2023.



Thanks

Do you have any questions?